

InstaCalc: The Road to Product–Market Fit

Why the daily users are the signal, seven testable hypotheses about what they actually come back for, what they love enough to double — and a strategy that has been argued against before being argued for.

CORE THESIS, HELD LOOSELY

InstaCalc's drive-by traffic is a funnel, not a product. The product is whichever job makes a small cohort return **daily** — and the honest truth is we don't yet know precisely which job that is. This document names **seven candidate daily jobs and the behavioral fingerprint that identifies each one**, isolates the qualities users most plausibly love, stress-tests the "own the maker" strategy against its strongest objections, and converts every major bet into a **hypothesis with a test, a threshold, and a kill criterion**.

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01 Executive summary

InstaCalc has been alive for nearly twenty years and has the one asset most products chasing product–market fit would kill for: **a cohort that returns daily without being asked to**. They build gaming calculators. They keep budgets, quotes, and life-math in it. They describe their usage as "managing their calc life." Version 1 of this document treated those facts as a settled diagnosis and prescribed a strategy. This revision does something more useful: **it treats them as evidence in an open investigation, attacks its own conclusions, and converts every bet into a test**.

What we know, what we assume, what we must find out

- **We know** the core interaction is loved by somebody: instant results as you type, variables that cascade, a share link in one click with no login. We know daily users exist and that a public corpus (1,000+ calcs, including gaming) grew organically.
- **We assume** — plausibly but without proof — that gaming theorycrafters are the wedge, that makers want profiles and forks, and that "calc life management" means a library product. Each assumption gets a test and a kill criterion in §13.
- **We must find out why the daily users actually come back**. "Daily use" is seven different possible jobs — a living dashboard of owned calcs, a default scratchpad habit, a template re-run ritual, a work tool, a gaming daily loop, a thinking medium, a running log — and *each implies a different product*. The good news: each job leaves a distinct fingerprint in usage logs, so two weeks of instrumentation plus one intercept question can tell us which tribe is biggest. That diagnosis, not this document, is the real strategy (§5).

The corrected LLM claim

Version 1 said LLMs are "eating the scratchpad job." That was half right. **LLMs eat one-shot answers and the search funnel** ("15% tip on \$84" never reaches a website anymore). **They do not eat the iterative what-if loop** — change one number, watch ten downstream results move, change it again — because chat has latency, no visible model state, and untrusted arithmetic. You can't easily go back and forth. The iterative scratchpad is therefore not a dying segment but **a defensible strength and a marketing angle**: InstaCalc is the back-and-forth surface a chatbot can't be. What LLMs do take is discovery — which is why distribution must come from communities, embeds, and shared links rather than search (§8, §11).

The strategy, post-red-team

"Own the maker" survives adversarial review, but humbled and re-sequenced:

1. **Diagnose the daily users first (weeks 1–4)**. Instrument seven fingerprints, run the Sean Ellis survey, ask daily users one question at open: "What did you come to do today?" Build for the biggest lovable tribe — don't average the tribes.
2. **Double what is loved before adding what is absent**. The love-objects are almost certainly some mix of *instantness, aliveness, zero ceremony, and legibility* (§6). Every roadmap item must protect these; several should amplify

them (global hotkey, live sliders everywhere, "update-in-place" dailies). A growth strategy that dilutes the love kills the patient to cure him.

3. **Ship Fork — it's utility, not social.** One click from any public calc to your own editable copy, with visible lineage. Highest leverage feature; low risk even if the network thesis is wrong, because forking is useful to a single user.
4. **Demote gaming from "the wedge" to "the first experiment."** The evidence for gaming-as-wedge is a gallery category and founder observation — real but thin. Seed three game communities with founding creators, measure views and forks for 30 days, and let the fork rate decide (kill criterion in §13). If it fires, pour on; if not, gaming was distribution, not identity — and the daily-job diagnosis picks the wedge instead.
5. **Social features only if pulled.** Profiles ship as attribution pages (a byline, not a feed). Follows and notifications wait for demonstrated fork demand. A quiet tool that bolts on community risks an empty mall and — worse — alienating the people who love it precisely because it's theirs and it's calm.
6. **Monetize power and organization, not participation** — with one adjustment from red-teaming: don't make *privacy* the headline Pro feature, or the most devoted users will take the corpus private. Charge for folders, data tables, deep history, clean embeds; keep a generous private allowance free (§12).

NORTH-STAR METRIC (UNCHANGED) AND THE DECISION CADENCE (NEW)

Weekly Returning Calc Owners (WRCO) — people who open or edit a calc they created or forked in a given week. New in v2: a hypothesis register (§13) where every major bet — gaming wedge, fork demand, daily-job identity, willingness to pay, AI referral — carries a metric threshold and an explicit kill-or-scale decision, most within 90 days. The strategy is allowed to be wrong; it is not allowed to be unfalsifiable.

02 Where InstaCalc stands today

Assets — what is already true

- **A genuinely differentiated core interaction.** Results as you type, variables that cascade, natural language, units and currency, Excel-style functions — and a public share link in one click with no login. That last part is still rare in 2026, and it is the seed of every growth loop in this document.
- **Longevity and trust.** Covered by TechCrunch in 2006; still alive, still fast. A twenty-year-old domain with organic links is an asset most startups cannot buy.
- **An existing UGC corpus.** A gallery of 1,000+ public calcs across business, cooking, **gaming**, and education. The supply side of a marketplace already exists in embryo.
- **Surface area for habit.** Chrome extension (Cmd+I), iOS app with sync, offline mode, command bar, presentation mode, sliders and dropdowns, revision history. Much of the "daily driver" product is already built — it is under-narrated, not missing.
- **The BetterExplained halo.** A large audience that already trusts the author on "math, explained intuitively." InstaCalc is the interactive twin of that promise, and the cross-pollination channel is essentially free.
- **Recent product energy pointed the right way.** Calculus support, previous-row references, dashboard, revision history — the community discussion themes show a tool being sharpened for people who *keep* calcs, not just people who compute once.

Signals — the evidence hiding in plain sight

Observation	What it means — and what it doesn't yet prove
People build gaming calcs	A community with intense, recurring math needs self-selected InstaCalc without being marketed to — the strongest kind of PMF lead. <i>Not yet proven:</i> that this is central theorycrafting rather than casual side-calcs, or that the segment will fork and spread. That's Hypothesis H4 (§13), not a fact.
Some people visit daily	A retention curve that flattens above zero — the PMF shape. Most calculator sites have zero true retention. <i>Not yet known:</i> what job drives the visits. Seven candidates, seven fingerprints (§5).
They "manage their calc life" in it	Users invented a job description the product doesn't officially offer — no real home screen, folders, or identity. They do this job <i>despite</i> the product. Build it and the job gets easier and stickier — <i>if</i> the daily job is library-shaped (H2 tests this).

The honest gap

- **No measured funnel.** The most important gap. Without cohort analytics and a PMF survey, "some people visit daily" stays an anecdote rather than a number that can be diagnosed and grown.
- **No maker identity.** A calc is an anonymous URL — no byline, no "more by this author," no reason a maker's audience becomes InstaCalc's audience.
- **No network structure.** Calcs can be shared but not visibly forked, followed, or collected. Each viral moment is a dead end instead of a graph edge.
- **Weak discovery.** The gallery exists but without usage ranking, category landing pages, or freshness signals.
- **No revenue engine.** The most devoted users have no way to vote with their wallets, and the product has no compounding budget for its own growth.

The engine is a decade ahead of the chassis. InstaCalc computes like a modern product and distributes like a 2006 widget — and measures like neither.

03 What product–market fit actually means here

"PMF" gets used loosely. For a horizontal free tool with twenty years of ambient traffic, it needs a precise definition, or the traffic itself will masquerade as fit.

Three tests, in order of rigor

1. **The retention test (the real one).** Plot cohort retention: of the people who create a calc in week 0, what fraction are back doing anything meaningful in weeks 4, 8, 12? PMF is the curve *flattening at a healthy level* rather than decaying to zero. The daily users say a flat asymptote exists — the job is to find out how high it is and for whom.
2. **The Sean Ellis test (the leading indicator).** Ask users who've used the product at least twice in the last two weeks: "*How would you feel if you could no longer use InstaCalc?*" If $\geq 40\%$ answer "very disappointed," the segment has PMF. Below 40%, the gap analysis (§13) tells you what to build. Cheap, fast, and the backbone of Superhuman's public PMF engine.
3. **The pull test.** Do users do unpaid work for the product — write guides, request features in public, build things you didn't anticipate? Gaming calc makers building for their communities is exactly this. Pull exists; it is narrow.

The crucial reframe: PMF is per-segment, per-job

A horizontal tool never has "PMF" in general — it has PMF *with someone, for some job*. Slack had it with small tech teams before enterprises; Notion with personal-wiki nerds before companies. The question is never "does InstaCalc have PMF?" but "**which segment already exhibits fit-shaped behavior, for which job — and can the product double down without losing what they love?**" The second clause is new in v2 and it is load-bearing: §10 examines the real possibility that growth features could damage the very qualities that make the daily users daily.

DEFINITION ADOPTED FOR THIS DOCUMENT

InstaCalc has product–market fit when **a named segment shows $\geq 40\%$ "very disappointed" on the Ellis survey, a flattening D30 retention curve, and a growth loop where existing users' calcs bring in next month's new makers without paid acquisition.** All three, for one segment, before generalizing.

04 The two products hiding inside InstaCalc

Every strategic confusion about InstaCalc dissolves once you see that it is two products sharing one codebase:

	Product A: The Scratchpad	Product B: The Calc Library
Job	"Think through this math, right now, back and forth, faster than a spreadsheet."	"Keep my numbers alive, organized, and shareable."
Session	90 seconds to 20 minutes; often unsaved	Recurring; named calcs; edited over months
Audience	Everyone, occasionally (huge, shallow) — plus some daily habituals	Makers & their communities (small, deep)
Retention	Low in aggregate — but the habituals exist (see \$5, D2)	Flattening — the PMF shape
Threats	LLMs take the <i>one-shot</i> slice and the search funnel; they cannot do the iterative what-if loop	Google Sheets inertia; abandoned bespoke fan sites
Strategic role	Top of funnel AND the core of what's loved. Keep sacred, keep free, make faster.	The compounding layer. Build, measure, monetize.

The LLM question, answered precisely (v2 correction)

Version 1 wrote off Product A too quickly. The accurate picture has three parts:

- **What LLMs take: one-shot answers and discovery.** "15% tip on \$84," "230 km in miles" — no iteration, answered in the chat box or omnibox the user is already in. This was the bulk of anonymous drive-by calculator traffic, and it also erodes the referral path: people who would have searched "compound interest calculator" now ask an assistant and never see a tool. The *funnel* shrinks even where the tool is better.
- **What LLMs can't take: the back-and-forth.** The core InstaCalc loop — tweak a variable, watch every downstream number recalculate instantly, tweak again — is structurally bad in chat: re-prompt latency, model state smeared across scroll history instead of visible as an editable structure, and probabilistic arithmetic that users rightly don't fully trust for real stakes. **You can't easily go back and forth with a chatbot.** For exploratory math — quotes, scenarios, "can I afford this" — chat is not a substitute; it is a worse tool. This is a durable strength and should be said out loud in positioning: *the what-if surface a chatbot can't be.*
- **The watch item: generated ephemeral widgets.** Chat surfaces increasingly spawn small interactive artifacts inline. That's the version of the threat that could eventually touch the iterative loop. But it produces *disposable* artifacts — unversioned, unowned, unshared, unmaintained. The defense is Product B: kept, named, linked, community-trusted calcs. And the flip side is an opportunity: assistants can't *be* the interactive surface, but they

can *link to one* — which only works for products whose artifacts are URLs. That is InstaCalc's whole DNA (§8, H8 in §13).

"Managing my calc life" is the user-invented name for Product B. When users name your product before you do, ship the product they named — after checking which product they actually meant (§5).

05 The daily-use mystery: seven hypotheses

Some people use InstaCalc every day. This is the most valuable fact about the product, and today it is unexplained. "Daily use" is not one behavior — it is at least seven distinct jobs, and **each implies a different product to build**. Averaging them would produce a product for nobody. The right move is to identify the tribes, size them, and build for the biggest one that is also lovable and reachable.

Each hypothesis below comes with the *behavioral fingerprint* that identifies it in usage logs — so this is a diagnosis to run, not a debate to have.

Hypothesis	What daily use looks like	Fingerprint in the logs	If it dominates, lean in with...
D1 Living dashboard	They re-open the same 2–5 owned calcs and update the numbers: net worth, budget, business metrics, a savings goal. InstaCalc as a personal numeric cockpit.	High same-calc reopen rate; edits change <i>values</i> , not formulas; calcs are old and long-lived.	Pinned home screen; "update in place" UX; deltas since last visit ("net worth +2.1%"); mobile/home-screen widget; gentle history charts of a value over time.
D2 Default scratchpad	InstaCalc is their reflexive math surface — Cmd+I, type, think, close. Different (often unsaved) math every day; the habit is the product.	Many short sessions on blank calcs; low save rate; entry via extension/hotkey/bookmark; little reuse.	Speed obsession (cold-load budget, offline default); global hotkey everywhere; optional new-tab page; auto-kept "recent scratchpads" history so throwaway work is silently recoverable.
D3 Template re-runner	Same <i>structure</i> , new inputs, over and over: freelancer quotes, recipe scaling, materials estimates, loan comparisons for clients.	Duplicate-calc events; repeated edits confined to input lines; share/export after each run.	"Run again" mode: saved template + clean input form; export a filled run as PDF/link; a personal template shelf.
D4 Embedded work tool	Part of a job workflow — engineering conversions, finance sanity checks, ops math. Used because it's faster than opening Excel.	Weekday 9–5 usage curve; business/engineering vocabulary in calcs; desktop-dominant.	Reliability messaging, privacy, import/export; this tribe is the natural first Pro buyer.
D5 Gaming daily loop	Daily game rituals: market flipping margins, energy/stamina optimization, event math. The calc is checked or updated when the game resets.	Usage periodicity matching game reset times/patch days; gaming-category calcs edited near-daily.	Editable "live" cells for prices/rates; patch version notes; per-game hubs; this is the bridge to the gaming wedge (H4).

D6 Thinking medium

They think *in* InstaCalc the way writers think in a text editor — meandering, mixing prose and math, exploring decisions. The BetterExplained ethos, lived.

Long sessions; text mode usage; many edits, few shares; calcs with sentences in them.

Notebook affordances: headings, notes, folding sections; "explain this formula"; beautiful reading mode. Position as a *place to think in numbers* — the tagline already says it.

D7 Running log

A calc that grows by a row a day: weight, spending, workout numbers, hours billed. The math summarizes the log.

Monotonic row growth over weeks; date functions; single-calc loyalty.

Quick-append UX (one field, enter, done — especially mobile); a sparkline of any column; optional reminders.

How to run the diagnosis (three weeks, near-zero code risk)

1. **Instrument the fingerprints (week 1).** Log per-user: calc reopen rate, save rate, edit type (value vs. structure), entry point, session length, duplicate events, row-append patterns, time-of-day. Every hypothesis above resolves to two or three of these numbers.
2. **Ask one question at the door (weeks 1–3).** For users on their 3rd+ visit in a week, a single unobtrusive prompt: "Quick one: what did you open InstaCalc to do just now?" with the seven options in plain words. Self-report + fingerprint together beat either alone.
3. **Interview the modal tribes (weeks 2–4).** Five conversations per dominant fingerprint. The script is short: *Walk me through yesterday's use. What did you do before InstaCalc? What almost made you stop? What would you pay for — and what would offend you to be charged for?*

WHY THIS SECTION IS THE REAL STRATEGY

Every downstream decision branches on the diagnosis. **If D1/D7 dominate**, the product is the personal dashboard — widgets, deltas, quick-append — and monetization is privacy-and-organization. **If D2 dominates**, the product is omnipresence and speed — hotkeys, new-tab, instant history — and the library is secondary. **If D3/D4 dominate**, templates and Pro come first. **If D5 shows up strongly**, the gaming wedge gets independent confirmation before we've spent a dollar on it. **If D6 dominates**, positioning shifts from "calculator" to "thinking medium," which changes everything downstream. The tribes likely coexist; the strategy is to name the biggest lovable one and serve it unmistakably well — the others still benefit, because all seven leans share the same foundations (speed, autosave, home screen).

06 What people love — and how to double it

PMF strategy has two halves: close the gaps, and **amplify the magic**. The second half is routinely neglected because gaps are legible and magic is taken for granted. For a product with twenty years of quiet devotion, the magic is the asset. Four love-objects, inferred from the product's design and its users' behavior — each with a way to confirm it and a way to double it:

Love-object	What it feels like	Confirm it	Double it
Instant-ness	Results appear <i>as you type</i> . No run button, no cell references, no waiting. The tool moves at the speed of thought.	Ellis-survey verbatims: does the "fast/instant" word-cluster dominate "very disappointed" answers?	Treat latency as a feature with a budget: sub-second cold load, offline-first, global hotkey, optional new-tab page. Market it: " <i>faster than opening your calculator app.</i> "
Aliveness	Change one number and everything downstream moves. The model feels like a living thing, not a worksheet — a spreadsheet without the grid.	Do users who discover sliders retain better? Do verbatims mention "playing with" numbers?	Sliders on any variable in one keystroke; drag-to-scrub any number; sparklines beside results; make the recalculation <i>visible</i> (a subtle ripple). Aliveness is also the anti-LLM pitch: chat can't do this.
Zero ceremony	No login, no filename, no template picker, no upsell. You are computing within one second of arriving. The tool disappears.	Funnel data: what % of first-time visitors compute within 10 seconds? Guard that number.	Constitutional rule: nothing may ever stand between a visitor and a working calc — not signup, not cookie banners styled like signup, not "new" badges. Autosave scratch work silently so ceremony never returns as "did you save?"
Legibility	A calc reads like reasoning: named variables, visible assumptions, math that looks like sentences. Your thinking stays inspectable — the BetterExplained DNA.	Interview question: "show me a calc you're proud of — why this way and not a spreadsheet?"	Beautiful reading/presentation mode; "explain this line"; shared calcs that look like short articles with live numbers. Legibility is why a shared InstaCalc beats a shared spreadsheet <i>as an artifact</i> .

The doubling principle

Superhuman's insight, applied honestly: after the survey, take the users who would be *very disappointed* to lose InstaCalc, read their own words for the main benefit, and **spend roughly half of all product effort making that benefit more extreme** — the other half closing the gaps that hold back the sympathetic middle. For InstaCalc the likely translation: half of effort on speed, aliveness, and the daily-job winner from §5; half on the chassis (fork, home screen, discovery). And a standing rule from the red team (§10): **any feature that would add ceremony, slow the**

first calc, or clutter the editor is rejected regardless of strategic appeal. The love is the moat; the network is the amplifier — never the reverse.

People don't love InstaCalc because it has features. They love it because it's the one place where numbers keep up with their thoughts. Every strategic move either sharpens that or dulls it.

07 Segment deep dive: who retains and why

Segment 1 — Gaming theorycrafters (confidence: promising, unproven)

Every game with numbers breeds a priesthood of players who do the math for everyone else: damage calculators, drop-rate and pity-timer calcs, crafting-profit sheets, build optimizers, XP-per-hour tables. This exists for every persistent game economy — MMOs, gacha, idle/factory games, tabletop.

Why they are underserved

- **Google Sheets is the incumbent, and it's miserable for this:** view-only links, "File → Make a copy" friction, brutal on mobile where players actually read Discord and Reddit, ugly embedded, anonymous and unforkable in any visible way.
- **Bespoke calculator sites are the other incumbent, and they die:** a fan builds a beautiful calc; the game patches; the fan gets a job; the community's institutional math evaporates.
- **Independent wikis need embeddable interactive tools** as communities migrate off legacy wiki farms — a slot InstaCalc embeds fit exactly.

Why they could be the perfect wedge

- **Built-in distribution:** one calc in a subreddit or Discord reaches thousands of players; some become makers.
- **Built-in retention:** patches change numbers; changed numbers mean updates and return visits — a recurring engagement clock most products spend millions to simulate.
- **Feature fit already exists:** sliders and dropdowns are precisely the "try your own gear score" interface; forking matches remix culture; identity matters to reputation-driven theorycrafters.

The adversarial caveat (new in v2)

The evidence is a gallery category and founder observation — enough to justify an experiment, not a bet-the-company wedge. Serious theorycrafting is data-heavy (large lookup tables) where Sheets is genuinely strong, and elite theorycrafters earn reputation from bespoke sites. The honest position: **gaming is Hypothesis H4 with a 30-day test and a kill criterion (§13), not a conclusion.** Note also that §5's D5 fingerprint gives an independent read: if gaming daily-loops are already a top tribe among daily users, confidence rises before we spend anything.

Segment 2 — Daily-numbers people (confidence: high that they exist; job mix unknown)

The people running their lives in InstaCalc: budgets, mortgage and FIRE scenarios, freelancer quotes, fitness macros, unit-heavy hobbies. Quieter than gamers, but they are the ones who visit *daily* — and §5 exists to find out which of the seven jobs they're doing. They are the natural first buyers of a Pro tier, because organization and depth are exactly what one pays for.

Segment 3 — Embedders: bloggers, educators, niche-site owners (confidence: market proven by others)

People who want an interactive calc *inside* their content. Omni Calculator proved the demand side with staff-built calcs drawing tens of millions of visits; Calconic and ConvertCalculator proved businesses pay \$5–50/month for embeds. InstaCalc's angle: the embed is free and instant, and readers can *fork* it — an interactivity story neither the SEO farms nor the form-builders can match. Every embed is a permanent, high-intent acquisition surface that AI search cannot intercept.

Segment prioritization

Segment	Passion	Distribution built-in	Willingness to pay	Role in strategy
Gaming theorycrafters	Extreme	Yes — communities	Modest (pride, flair)	First experiment for the network loops (H4)
Daily-numbers people	High (private)	Low	High — organization & depth	Retention core + first revenue; \$5 names their job
Embedders / creators	Moderate	Yes — their sites	High — proven market	Amplifier + second revenue

One product serves all three: the maker-centric calc library. The segments differ in which loop they power — and in how much we currently know about them.

08 The competitive map and the LLM question

Player	What they own	Why they won't take InstaCalc's ground
Desmos	Math <i>education</i> : graphing, classrooms, standardized tests	Institutionally anchored to curriculum. Doesn't care about a drop-rate calc or your freelance quote sheet.
Soulver / Numi / Calca	The beautiful <i>local</i> notepad calculator (paid Mac/iOS apps)	Deliberately personal and offline. No sharing, no URLs, no community, no embeds. They validate the interaction model while ceding the network entirely.
Omni Calculator	SEO for "X calculator" queries; thousands of <i>staff-authored</i> calcs	Closed supply — users consume, never create or fork. They proved demand is enormous; they structurally cannot do UGC, community, or personal libraries. (And their search channel is the one AI answers erode fastest.)
Calconic / ConvertCalculator / Calculoid	Paid B2B embeds: pricing quotes, lead-gen forms with checkout	Sales-tool DNA, per-impression pricing. Zero consumer joy, zero community. They proved businesses pay monthly for embedded calcs.
Google Sheets	Default gravity for any shared numbers	The real competitor. Beaten not on power but on <i>experience of the shared artifact</i> : instant, mobile-friendly, forkable, legible vs. view-only spreadsheet sprawl.
LLM chat	One-shot answers and the discovery funnel	Cannot do the iterative what-if loop (latency, no visible model state, untrusted arithmetic — you can't go back and forth). Can generate ephemeral widgets, but not kept, named, community-trusted, maintained artifacts. Court it as a <i>referrer</i> : assistants can link to interactive calcs, and only URL-native products can receive that traffic.

THE WHITESPACE

Nobody owns the community-made calculator — the fan-made DPS calc, the forked FIRE model, the brewing calc a subreddit swears by. Desmos won't (education), Omni can't (closed supply), Soulver won't (local-only), Calconic won't (B2B), and LLMs produce answers and ephemera, not maintained artifacts. InstaCalc has the engine, the one-click-share DNA, the gallery seed, and twenty years of trust. The position: *"GitHub for calculators"* in mechanics, *"the home for your numbers"* in feeling — plus, new in v2: *"the what-if surface a chatbot can't be"* in positioning against the AI tide.

A useful historical rhyme: CodePen for code snippets, Figma Community for design files, GitHub for repos. In each case a utility became a durable network the moment artifacts got identity (authorship), lineage (forks), and a social

surface (profiles, counts, discovery). Calculators are late to this pattern — that's the opportunity. The red team (\$10) will test whether the pattern actually transfers.

09 The strategy: own the maker

Positioning

InstaCalc is where calculations live. Make a calc in seconds, play with the numbers in real time, share it with a link, fork anyone else's, and keep your whole calc life in one place — free, instant, no login to view or try.

Three constitutional commitments:

1. **Viewing, sharing, and forking are free forever.** The loop is sacred; never tax it.
2. **A shared calc must be the best possible artifact of its kind** — faster, prettier, more legible, and more usable on a phone than a Google Sheets link, every single time.
3. **The blank calc stays sacred.** Nothing may slow, clutter, or interrupt the core type-and-see experience — no matter how strategically appealing. (Added by the red team, §10, Attack 3.)

Product priorities (re-sequenced in v2)

Tier 0 — Diagnose before building (weeks 1–4; §5 and §13)

- Instrument the seven fingerprints; run the Ellis survey; run the at-the-door intercept; interview the modal tribes.
- Tier 2 is written in pencil until this returns.**

Tier 1 — Utility-safe chassis (valuable under every hypothesis)

- **Fork.** A prominent "Fork this calc" on every public calc: one click → your own editable copy, with visible lineage ("forked from ...", "34 forks"). Highest-leverage feature in the document — and crucially, *useful to a single user even if the network thesis fails* (it's "make this mine," not social).
- **The Library (home screen).** Your calcs: pinned, foldered, searchable, recently edited. The literal "manage your calc life" feature. Extension and iOS open here.
- **Great unfurls.** Rich previews so a calc pasted into Discord, Reddit, Slack, or iMessage shows title, author, and a live-looking preview. For community distribution the unfurl is the ad. Cheap to ship; ship first.
- **Maker attribution.** Profiles as modest byline pages — a name, an avatar, a list of public calcs. Not a feed, not followers. Social mechanics wait for evidence (Tier 3).

Tier 2 — Double the love + serve the winning daily job (from §5/§6 results)

- **Speed and aliveness work** regardless of tribe: cold-load budget, drag-to-scrub numbers, one-keystroke sliders, sparklines.

- **Then the tribe-specific lean:** dashboard deltas and widgets (D1), global hotkey and new-tab page (D2), templates with input forms (D3), quick-append and column sparklines (D7), live-cell prices and version notes (D5), notebook affordances (D6).

Tier 3 — Network and discovery (gated on Tier 1 evidence)

- **Version notes & freshness labels** ("updated for patch 2.4") — gate: fork demand confirmed (H3).
- **Follow / notify** on calcs and makers — gate: repeat-viewer behavior observed on popular calcs.
- **Gallery v2:** usage-ranked category pages, trending, per-game hubs — gate: enough active supply that curation shows a live corpus, not a quiet mall.
- **Data tables** (paste/import CSV, reference from formulas) — gate: gaming test (H4) survives, or D3/D4 tribes demand it.
- **AI-assisted creation** ("describe your calculator" → editable draft) — collapses the maker learning curve; also the honest internal use of AI.

Why this wins

Engines can be copied; **corpora, habits, and communities cannot**. If the bets pay, the eighteen-month moat is: thousands of community-maintained calcs interlinked by forks, ranked by real usage, embedded across wikis and blogs, authored by people whose reputations live on their bylines — wrapped around a core interaction that stayed fast and beloved. A competitor can clone the calculator in a quarter; they cannot clone the graph or the habit.

10 Red team: the strongest objections, argued honestly

Each attack below is the steelman — the version a smart skeptic would actually make — followed by the conclusion it forces. Several changed the strategy; that is the point of the exercise.

ATTACK 1 — "THE GAMING WEDGE IS FOUNDER-ROMANTIC, NOT EVIDENCE-BASED."

A gallery category and anecdotes prove casual gaming calcs exist, not that theorycrafters will adopt. Serious theorycrafting is data-heavy — thousand-row item tables, simulation, spreadsheet lookups — exactly where Sheets is strong and InstaCalc is weak. Elite theorycrafters build bespoke sites *because* the site is the reputation. You may seed three communities and get polite upvotes and no forks.

Conclusion: Demoted from "the wedge" to "the first experiment." Spend 30 days and near-zero engineering (seed calcs + founding creators + measure) before any gaming-specific roadmap. Kill criterion lives in §13 (H4). Target the *casual-to-mid* theorycrafting layer (quick build checks, market margins, daily-efficiency math) where Sheets friction hurts most — not the simulation elite. Watch §5's D5 fingerprint for independent confirmation before spending.

ATTACK 2 — "SOCIAL FEATURES ON A QUIET TOOL = AN EMPTY MALL."

Bolted-on community is the most common failure mode in tools: profiles nobody visits, follows nobody follows, a feed of tumbleweed that makes a healthy product look dead. Worse, the people who love InstaCalc may love it *because* it is quiet, personal, and theirs — a calm corner of the internet. Injecting engagement mechanics could repel the very cohort with PMF.

Conclusion: Fork ships because it is *utility* (one click to "make this mine") — valuable even to a user who never socializes. Profiles ship as *bylines*, not destinations. Follows, notifications, and anything feed-shaped wait for demonstrated fork demand (H3). The home screen is always *your* calcs, never a feed. Empty-mall risk is also why seeding concentrates density in three communities rather than spreading thin.

ATTACK 3 — "YOUR GROWTH PLAN COULD KILL THE THING PEOPLE LOVE."

Daily users chose a tool with no login walls, no upsells, no clutter, and no audience watching them. Every strategic addition — banners for Pro, fork counts, gallery promotions, AI assistants — is a tax on the calm. Products regularly destroy PMF they already had by optimizing for the users they don't yet have.

Conclusion: Adopted as a constitutional rule (§9): the blank calc is sacred — nothing may slow, clutter, or interrupt type-and-see. All network surface lives *around* calcs (share pages, gallery, profiles), never *inside* the editor. Track the §6 guard-metrics (time-to-first-calc, cold load) in CI like tests. Re-run the Ellis survey after each major release; a falling "very disappointed" % among existing dailies is a rollback signal, not a shrug.

ATTACK 4 — "UGC SEO IS A CLOSING WINDOW, NOT AN OPEN PLAYBOOK."

Omni's playbook worked in a world where "mortgage calculator" queries reached websites. AI answer boxes now absorb exactly these queries; search engines penalize thin UGC pages; and a corpus of user-made calcs is, by search-quality standards, mostly thin pages. Betting a growth loop on 2015-era SEO is nostalgia.

Conclusion: The SEO loop is demoted to opportunistic (§11): make calc pages crawlable and well-titled because it's nearly free, curate category hubs, and expect little. The durable channels are the ones AI can't intercept — communities, embeds, and direct shares. Meanwhile, *court the successor channel*: make calc pages machine-readable (clean metadata, structured data) so AI assistants can cite and link interactive calcs. That's H8 — small bet, asymmetric upside.

ATTACK 5 — "MONETIZING PRIVACY STARVES THE PUBLIC CORPUS."

The network strategy needs public calcs; the Pro tier charges for private ones. Your most devoted users — the likeliest subscribers — are then systematically pulled out of the commons. You'd be selling the network's food supply.

Conclusion: Restructured pricing (§12): a generous free private allowance (private-by-obscurity links plus a handful of truly private calcs); Pro charges for *organization and power* — unlimited folders, deep history, data tables, clean embeds, exports — rather than privacy as the headline. Monitor the public:private ratio of new calcs as a standing health metric.

ATTACK 6 — "THIS PLAN IS THREE COMPANIES' WORTH OF WORK FOR ONE PERSON."

Diagnosis, chassis, love-doubling, community seeding, monetization, AI channels — a solo founder shipping all of it ships none of it well. The strategy's real enemy is WIP.

Conclusion: Hard WIP limit encoded in the 90-day plan (§13): exactly *three* engineering deliverables in the first quarter (instrumentation, unfurls, fork) — everything else is measurement, conversation, and seeding, which communities and founding creators carry. Tier 2 and 3 are explicitly gated on evidence, which is also a scheduling device: the data decides what *not* to build.

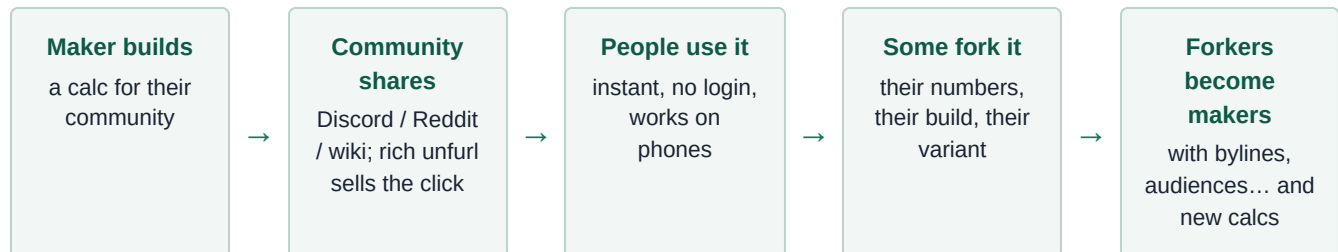
What survived unchanged

The two-product analysis (§4), the maker-centric direction, fork as the keystone feature, community-over-search distribution, and the refusal to paywall the loop. The red team sharpened sequencing and pricing; it did not find a better strategy — provided the §5 diagnosis confirms that a library-shaped or maker-shaped tribe actually dominates daily use. If it doesn't, §5's branch table is the contingency plan, and that is exactly how it should be.

11 Growth loops

Three primary loops and one demoted one. Each is measurable; none require paid acquisition.

Loop 1 — The Fork Loop (the flagship, gated on H3/H4)



Measure: share-viewer → fork conversion; forks per top-100 calc; new makers per month attributable to a fork.

Accelerant: the patch clock — every game update re-fires the loop for free.

Loop 2 — The Embed Loop (creator-powered)

Blogger or wiki embeds a calc → readers interact → footer link ("fork it") converts a slice into users → some embed on *their* sites. AI search cannot intercept an embed; this loop ages well. **Measure:** active embeds; click-through per embed; embed → creation rate.

Loop 3 — The Patch Loop (retention as growth)

Game patches → maker updates calc with a version note → followers notified → audience returns en masse → renewed sharing restarts Loop 1. No general-purpose competitor can copy this because it lives in the domain, not the product. **Measure:** return-visit spikes around major patches of the top games in the corpus. **Gate:** exists only if H4 survives.

Loop 4 — SEO (demoted to opportunistic; see Red Team, Attack 4)

Crawlable calc pages, honest titles, curated category hubs — because it's nearly free, not because it's the plan. The forward-looking replacement: **machine-readable calc pages that AI assistants can cite and link** (H8). An assistant that answers a mortgage question and links "try the interactive version" sends traffic only products with URL-native artifacts can receive.

Seeding the wedge experiment (the cold-start work for H4)

1. **Pick three games** with active economies, frequent patches, and spreadsheet-heavy communities (one MMO, one gacha/live-service, one idle/factory).
2. **Recruit one founding theorycrafter per game** (paid; day-one byline). Build 3–5 exemplary, patch-current calcs per game, targeting the casual-to-mid layer, not the simulation elite.

3. **Launch inside the community** — subreddit, Discord, wiki embed. The pitch is the calc, never the platform.
4. **Read the fork rate and decide** per the H4 criterion (§13). Pour on, or conclude gaming is a channel rather than the wedge — both are useful answers.

12 Monetization: who pays, for what, and one real tension

Principle: **charge for power and organization** — **never for participation, and not primarily for privacy** (the red team's correction: pricing privacy as the headline would pull the most devoted users' calcs out of the public corpus the network needs).

Tier	Price	What's in it	Who buys, and why
Free	\$0	Unlimited public calcs, sharing, forking, viewing, embeds (with footer), basic library, a handful of private calcs	Everyone. The network. Non-negotiable generosity — including enough privacy that free users never feel surveilled.
Pro	~\$6/mo or ~\$48/yr	Unlimited folders & private calcs, deep version history, data-table import, exports (PDF/image/CSV), template forms, profile flair	The winning \$5 tribe: dashboard-keepers and template re-runners buy organization and depth; theorycrafters buy data tables and pride.
Creator / Site	~\$15–29/mo	Unbranded embeds, embed styling/themes, embed analytics, custom subpath, multiple embeds	Bloggers, wikis, educators, small businesses — the market Calconic already proved at similar prices, with a worse product.
Later: Teams	TBD	Shared workspaces, permissions	Only if organic pull appears. Do not build speculatively.

Guardrails

- **Nothing a loop touches is ever paywalled:** viewing, sharing, forking, the first calc, the blank calc.
- **No ads on maker pages or embeds.** At most, one tasteful self-promo slot on anonymous drive-by pages.
- **Test willingness to pay before building the tier** (H6, §13): a pricing page with intent capture, or a founder-letter lifetime deal (~\$99) to the mailing list — twenty years of goodwill converted into capital, testimonials, and a real demand signal.
- **Watch the public:private ratio** of new calcs monthly; if Pro shifts it hard private, rebalance the tier toward power features.
- **Scale check:** 1,000 Pro subscribers $\approx$ \$48–72k/yr — a healthy indie business that funds everything here; 10,000 $\approx$ a real company. Neither requires beating anyone; both require the loops to turn.

13 The hypothesis register & 90-day test plan

Every major bet, with its test, threshold, and the decision each outcome forces. Thresholds are calibration points, not gospel — but a bet that misses badly gets killed, not renegotiated.

#	Hypothesis	Test	Threshold	Decision
H1	A "very disappointed" segment already exists (love is real and measurable)	Ellis survey to all 2x-in-2-weeks users; segment responses by use case	≥40% VD in at least one nameable segment	Pass: that segment's verbatims set positioning & Tier 2. Miss: mine the "somewhat" cohort's gap list before building anything else.
H2	Daily use is dominated by a library-shaped job (D1/D3/D7) rather than pure scratchpad habit (D2)	Fingerprint instrumentation + at-the-door intercept (\$5), 3 weeks	≥40% of daily users show an owned-calc reopen pattern	Pass: Library/home-screen is Tier 2's center. Miss (D2 wins): pivot Tier 2 to omnipresence — hotkey, new-tab, instant history — and revisit the maker thesis honestly.
H3	Fork demand exists (viewers want to make calcs theirs)	Ship Fork; measure for 60 days on public calcs	≥3% of share-page visitors fork; ≥15% of forkers later create or meaningfully edit	Pass: unlock Tier 3 (follows, freshness, gallery v2). Miss: keep fork as quiet utility; cancel social roadmap; network thesis downgraded.
H4	Gaming communities adopt seeded calcs (the wedge experiment)	3 games × 3–5 founding-creator calcs, launched in-community; 30 days	In ≥1 of 3 games: a seeded calc reaches ~1k+ views and ~25+ forks	Pass: scale to 8–10 games, build data tables & patch notes. Miss: gaming is a channel, not the wedge; the \$5 winner takes its budget.

H5	The love-object is instant-ness/aliveness (not, e.g., organization or sharing)	Code the VD verbatims from H1 by word-cluster	A single cluster ≥50% of VD responses	Pass: that cluster gets half of product effort (\$6) and the marketing headline. Split verdict: love differs by tribe — market per-segment.
H6	Devoted users will pay	Pricing page with intent capture, or founder-letter lifetime deal to the list	≥2% of actives show purchase intent, or ~100 lifetime buys	Pass: build Pro scoped to what buyers cited. Miss: delay Pro; revenue via Creator embeds instead.
H7	Embeds convert readers into users	Instrument the embed footer for 60 days	≥1% of embed viewers click through; ≥10% of those try a calc	Pass: invest in Embed v2 + Creator tier. Miss: embeds are brand, not funnel — deprioritize.
H8	AI assistants become a referral channel for interactive artifacts	Ship machine-readable calc pages (metadata, structured data); track assistant referrers quarterly	Nonzero and growing quarter-over-quarter	Pass: deepen (an API/feed assistants can cite). Miss: costs ~nothing; leave it on.

The 90-day plan (three engineering deliverables, everything else is learning)

Days 1–30 — Diagnose

- Ship **instrumentation** (deliverable 1): fingerprint events + the five-number dashboard (WRCO, new makers/wk, forks/wk, D30 maker retention, share-view → fork).
- Run the Ellis survey (H1, H5) and the at-the-door intercept (H2). Begin interviews with the modal tribes.
- Ship **rich unfurls** + **crawlable calc pages** (deliverable 2) — small effort, immediate distribution payoff, feeds H8.
- Recruit the three founding theorycrafters.

Days 31–60 — Build the keystone

- Ship **Fork with lineage** + **byline pages** + **Library v1** (deliverable 3 — one coherent release).
- Founding creators build seed calcs on their new bylines.
- Read H1/H2/H5 results; write Tier 2 in ink.

Days 61–90 — Test the wedge and the wallet

- Community launches in the three games (H4 clock starts).
- Pricing intent test (H6).
- Day 90: sit down with the register. Every hypothesis has data or a date. Decide: which tribe, gaming in or out, Pro scope, Tier 3 gates.

12-month milestones (if the bets pass)

Milestone	Target	Meaning
Ellis VD in the winning segment	≥ 40%	Segment PMF confirmed
WRCO growth	~10%/mo for 2 consecutive quarters	The daily job is compounding
Fork loop	Top-20 calcs avg ≥10 forks; ≥25% of new makers arrive via fork	The network is forming
Wedge breadth	8–10 games with an actively maintained hub (if H4 passed)	Repeatable playbook, not a one-off
Revenue	Pro live; 300–1,000 subscribers	The devoted can finally pay; the strategy self-funds

14 Risks and what *not* to do

Risk register

Risk	Reality	Mitigation
LLMs erode the funnel	One-shot math and "X calculator" searches increasingly never reach a website. The iterative what-if loop and kept artifacts remain defensible (§4).	Distribution via communities, embeds, and shared links — channels AI can't intercept. Court assistants as referrers (H8). Position explicitly as the back-and-forth surface chat can't be.
Growth kills the love	The daily users chose calm, speed, and zero ceremony; strategy adds surfaces.	Constitutional rule: the blank calc is sacred; network features live around calcs, never inside the editor. Guard-metrics in CI; re-survey after major releases; falling VD among dailies = rollback signal.
Spreadsheet gravity	Sheets is free, familiar, data-strong, and already where communities work.	Win on the experience of the <i>shared artifact</i> (instant, mobile, forkable, legible); add data tables so power users don't have to choose. Target the casual-to-mid layer where Sheets friction is highest.
Empty-network cold start	Profiles and fork counts look sad at low volume; social tumbleweed repels.	Fork-as-utility first; bylines not feeds; density in three communities before breadth; Tier 3 gated on H3.
Solo-founder bandwidth	The plan must run on one person plus community.	Three engineering deliverables per quarter, hard limit. Founding creators carry seeding; the register decides what not to build. Cut scope before cutting the loop.
UGC quality & misinformation	Wrong calcs erode the trust the brand runs on.	Curated gallery front; usage-ranked discovery; freshness labels; report link. Curation is editorial work — budget for it.
A well-funded copycat	The engine is cloneable in a quarter.	The corpus, fork graph, embedded footprint, habits, and maker reputations are not. Speed to network is the defense — which is why Tier 1 ships before monetization.

What *not* to do

- **Don't average the tribes.** Seven daily jobs exist; building a little for each produces a product for none. Diagnose, pick, commit.
- **Don't bolt on social before fork demand is proven.** Bylines yes; feeds, follows, and karma no — until H3 says otherwise.
- **Don't chase the B2B embed market head-on** — Calconic's war needs a sales motion. Take the creator slice with a better free product.
- **Don't compete with Desmos for education** — procurement and curriculum are a different company.
- **Don't keep adding engine depth as the growth plan.** The marginal function wins no new segment; the chassis and the love-doubling win several.
- **Don't bet on SEO.** Take what's free; build for the channels AI can't intercept.
- **Don't launch "the platform."** Launch calcs, inside communities, one at a time. Platforms are recognized in retrospect.
- **Don't paywall anything a loop touches.** Ever.

CLOSING

The daily users have already voted with the most honest currency there is: their habits. The revision's discipline is to stop guessing what they voted *for*. Three weeks of fingerprints and one survey will name the job; the love-objects tell us what to double; the register keeps every bet honest with a kill criterion; and the red team's rules keep growth from trampling the quiet, instant, alive little tool that earned the habit in the first place. *Think in numbers. Share a link. Keep your calc life here — and find out exactly why they already do.*